

Nguyễn Hồng Quân

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EDUCATION

[University of engineering and technology - VNU]

Bachelor of Science in Artificial Intelligence

Hanoi, Vietnam

2025 – Present

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Linear Algebra, Machine Learning.

PROJECTS

DealConnect | AI Startup & Investor Matching Platform | Python, LLMs, Google API

July 2026 – Aug 2026

- Architected an end-to-end matchmaking pipeline leveraging **Sentence-Transformers** embeddings and cosine vector similarity to semantically pair early-stage startups with domain-aligned investors.
- Integrated **Large Language Models (LLMs)** via structured prompting to extract key investment theses, ticket size parameters, and synthesize automated pitch summaries.
- Engineered automated scheduling workflows integrating **Google Calendar API** with OAuth 2.0 to detect reciprocal availability and dispatch meeting invites seamlessly.
- Optimized inference throughput with embedding caching and vectorized comparisons, accelerating matchmaking turnaround by **40%**.

Tic-Tac-Toe AI Engine | Minimax & Alpha-Beta Pruning | C++, SDL2, CMake, OOP

April 2026 – May 2026

- Engineered a modular, extensible game engine in modern **C++** implementing the **Minimax algorithm with Alpha-Beta pruning** for zero-sum adversarial decision making.
- Applied **SOLID principles and OOP design patterns** (Strategy, Factory) to decouple game state and evaluation logic from rendering layers.
- Supported multiple user interfaces including a lightweight Terminal CLI and an interactive **SDL2 graphical renderer** with configurable bot difficulty levels.
- Optimized state-space evaluation functions and recursive search depth to achieve sub-millisecond AI move decisions.

MicroTensor | NumPy Autograd Engine & MLP Framework | Python, NumPy, Math

Aug 2026 – Sept 2026

- Engineered a lightweight automatic differentiation engine from scratch in pure **NumPy**, implementing dynamic computation graphs and the chain rule for reverse-mode backpropagation.
- Developed a modular Neural Network API with fully connected layers, non-linear activations (ReLU, Sigmoid), and loss functions (MSE) to construct and train Multi-Layer Perceptrons.
- Validated the mathematical correctness of the engine by successfully training on non-linear datasets (e.g., Breast Cancer Wisconsin dataset) and cross-referencing gradient computations against **PyTorch**.

TECHNICAL SKILLS

Languages: C++, Python, SQL, HTML/CSS, Bash

AI / Machine Learning: Sentence-Transformers, LLM APIs (Gemini, OpenAI), Semantic Search, Scikit-learn, Pandas, NumPy

Systems & Build Tools: Linux, MSYS2, CMake, Ninja, VS Code, Git, GitHub

Concepts & Architecture: Object-Oriented Programming (OOP), Data Structures & Algorithms, Design Patterns, APIs

TECHNICAL SKILLS

Languages: Python, C++, HTML/CSS,

AI / Machine Learning: Sentence-Transformers, LLM APIs (Gemini, OpenAI), Vector Search, Scikit-learn, XGBoost, Pandas, NumPy

Tools & Frameworks: FastAPI, Google APIs, Git/GitHub, Docker, Linux, VS Code, CMake, Ninja

AWARDS & CERTIFICATIONS

Certifications: IELTS 7.5

Competitions: Vietnam AI Contest(VAIC)